

Indrashish Saha

CONTACT INFORMATION

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RESEARCH INTERESTS

Crystal Plasticity Finite Element, Fracture Mechanics, Damage Mechanics
Machine Learning, Data-driven Models, Surrogate Models

RESEARCH EXPERIENCE

Johns Hopkins University, Baltimore, USA

PH.D. THESIS

August 2021-Present

- Developing a microstructure dependent damage nucleation and growth model for materials under dynamic loading conditions using ML methods with specific applications to spall failure at extreme strain rates of $\sim 10^6 s^{-1}$.
- Developed cohesive zone insertion tool in Python to insert zero-thickness cohesive zone elements at every element boundary in a regular or irregular mesh for dynamic fracture applications in polycrystalline materials.
- Developed a cohesive zone coupled crystal plasticity (CP) model for the plate impact experiment (at strain rates of $\sim 10^6 s^{-1}$) of polycrystalline materials. The model incorporates cohesive zones at the grain boundaries to allow fracture of the microstructures at grain boundaries under dynamic loading. Additionally developed a data-driven model to predict the spall fracture in metals under shock loading using neural operators.
- Developed a data-driven model to predict the nonlinear plastic behavior of two-phase fiber-reinforced composites using 3D Convolutional layers in a U-Net structure. The model captures the loading and unloading of the microstructure in a given complex loading condition. The model generates the stress and strain maps of a given composite in a fraction of the time taken by traditional finite element methods.
- Advisor: Dr. Lori Graham-Brady, Professor, Department of Civil and Systems Engineering, JHU

Indian Institute of Science, Bangalore, India

M.TECH (RESEARCH) THESIS

December 2019-2021

- Worked on the statistical analysis of Acoustic Emission waveforms emitted from Steel fiber reinforced concrete (SFRC) under multiple loading condition and developed a machine learning method to characterize fracture processes in SFRC.
- Advisor: Dr. R. Vidya Sagar, Principal Research Scientist, Department of Civil Engineering, IISc Bangalore

EDUCATION

Johns Hopkins University, Baltimore, USA

PHD IN MECHANICS OF MATERIALS

2021-Present

- Department of Civil Engineering and Systems Engineering
- Current Cumulative GPA: **3.90 out of 4.**

Indian Institute of Science, Bangalore, India

M.TECH (RESEARCH) IN STRUCTURAL ENGINEERING

2019-2021

- Department of Civil Engineering
- Cumulative GPA: **8.60 out of 10.**

Jadavpur University, Kolkata, India

B.E. IN CIVIL ENGINEERING

2014-2018

- Cumulative GPA: **8.22 out of 10.**

PUBLICATIONS

- **Indrashish Saha** and Lori Graham-Brady (2026) Numerical and data-driven modeling of spall failure in polycrystalline ductile materials *Computer Methods in Applied Mechanics and Engineering* (Elsevier) [I.F. = 6.9]
- Ishan Khurjekar, **Indrashish Saha**, Lori Graham-Brady, Somdatta Goswami (2025) Enhanced accuracy through ensembling of randomly initialized auto-regressive models for time-dependent PDEs *arXiv* (Preprint) [Accepted]
- **Indrashish Saha**, Ashwini Gupta, Lori Graham-Brady, (2023) Prediction of local elasto-plastic stress and strain fields in a two-phase composite microstructure using a deep convolutional neural network *Computer Methods in Applied Mechanics and Engineering* (Elsevier) [I.F. = 6.9]
- **Indrashish Saha** and R. Vidya Sagar, (2022) Statistical analysis of acoustic emission avalanches generated during the compressive fracture process, and Mode I fracture process in cementitious composites. *International Journal of Fracture* (Springer) [I.F. = 2.7]
- **Indrashish Saha** and R. Vidya Sagar (2021) Classification of the acoustic emissions generated during the tensile fracture process in steel fibre reinforced concrete using a waveform-based clustering method, *Construction and Building Materials* (Elsevier) [I.F. = 7.4]

CONFERENCE PRESENTATIONS

- “Data-driven surrogate model for high velocity impact of polycrystalline metals” at *18th U.S. National Congress on Computational Mechanics*, 2025, Chicago, Illinois, USA.
- “Deep learning based accelerated high strain rate simulations for design of materials in extreme environments” at *16th World Congress on Computational Mechanics / 4th Pan American Congress on Computational Mechanics*, 2024, Vancouver, BC, Canada.
- “Deep learning based accelerated high strain rate simulations for design of materials in extreme environments” at *Society of Engineering Science (SES) - Annual Technical Meeting*, 2023, Minneapolis, USA.
- “Prediction of Stress Field in Fiber-Reinforced Composites Using 3D U-Net Under Cyclic Loading Conditions” at *10th International Conference on Multiscale Materials Modeling*, 2022, Baltimore, USA.

RELEVANT COURSEWORK

- Continuum Mechanics
- Continuum Plasticity
- Mechanical Properties of Materials
- Non-linear Optimization
- Deep Learning
- Finite Element Methods

TECHNICAL SKILLS

- Programming language: Fortran, Python, C++
- ML/DL libraries: PyTorch, TensorFlow, JAX
- Software: Abaqus (UMAT/VUMAT), MATLAB, FEniCS

ACHIEVEMENTS

- Received the **best paper award** at the Engineering Mechanics Institute (EMI) Conference in the ML in Mechanics paper competition, 2026.
- Received the **best poster award** at Hopkins Extreme Materials Institute (HEMI) day symposium, 2026.
- Received a **travel award** for the USNCCM-18 conference in 2025.
- Received the **P.S. Narayana Medal** for “Best M.Tech Research Thesis” at the **Indian Institute of Science, Bangalore** in 2024.
- Got **99.38 percentile** in **GATE** 2019 out of over 1.2 million candidates.